Instructions for reproducing the analyses compiling in 'Trophic interactions explain widely observed production and biomass scaling relationships' (Song, Atkinson, Talmy, Omta)

Anne Willem Omta

Below, there are brief technical instructions for reproducing the analyses presented in Figs. 2 & 3 of 'Trophic interactions explain widely observed production and biomass scaling relationships' (*Song et al.*, 2026). Please send an e-mail to anne.omta@case.edu, in case you encounter any problems.

Follow the AUTO07P manual (*Doedel and Oldeman*, 2009) to install the AUTO bifurcation analysis package. As a starting point for the bifurcation analysis, we used a steady state that we found using the program settle.f under Fig2/FindEq. Under Unix/Linux, this FORTRAN code is compiled with

f95 -o set settle.f

and the executable set is run as a background process with

./set &

Output of P and Z as a function of time is written to fort.70 and fort.80, respectively. The final steady-state values were then used as the input in the AUTO run file hybrid.f90 under Fig2 (P=U(1), Z=U(2)).

References

Doedel, E. J., and B. Oldeman (2009), *AUTO07P: Continuation and bifurcation software for ordinary differential equations*, Concordia University, Montreal.

Song, J., O. Atkinson, D. Talmy, and A. W. Omta (2026), Trophic interactions explain widely observed production and biomass scaling relationships, *submitted*.